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AI POWERED MULTIMODAL RESEARCH AND STUDY ANALYSIS FOR ENGINEERING STUDENTS

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for
the Course of*

CSA6502 – Generative AI and Large-Scale Model

to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL
INTELLIGENCE)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AI Powered Multimodal Research and Study Analysis for Engineering Students” has been carried out by J Karthik (192472136), V Rohan (192472227), Joanathan (192472229), M Chaitanya (192472047) under the supervision of Dr. J Nithisha and is submitted in partial fulfilment of the requirements for the current semester of the B.E. Computer Science and Engineering (Artificial Intelligence) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, J Karthik, V Rohan, Joanathan and M Chaitanya of the Computer Science and Engineering (Artificial Intelligence), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “AI Powered Multimodal Research and Study Analysis for Engineering Students” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 04/09/2026

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
J Karthik (192472136)	Multimodal Input & Document Processing module (Module 1)	Designed the ingestion pipeline and extraction routing logic; implemented OCR/layout analysis and Whisper-based transcription; implemented the confidence-flagging stage	Wrote and executed test cases for OCR accuracy (NED), ASR accuracy (WER), and low-confidence flagging (Section 4.3)	Chapters 3 and 4 (Module 1-specific sections), Section 5.3 Individual Reflection, and this Individual Contribution Statement	25%
V Rohan (192472227)	Multimodal Knowledge & Retrieval module (Module 2)	Designed the chunking strategy and vector-database schema; implemented embedding generation and top-K semantic retrieval; implemented cross-document linking	Wrote and executed test cases for chunking coherence, embedding coverage, retrieval relevance, and cross-document linking (Section 4.3)	Chapters 3 and 4 (Module 2-specific sections), Section 5.3 Individual Reflection, and this Individual Contribution Statement	25%

Joanathan (192472229)	Research & Study Analysis module (Module 3)	Designed the retrieval-augmented reasoning pipeline and consistency-checking logic; implemented summarisation, comparison, and grounded question-answering functions	Wrote and executed test cases for summarisation quality (BERTScore), comparison coverage, answer traceability, and multimodal consistency (LCI) (Section 4.3)	Chapters 3 and 4 (Module 3-specific sections), Section 5.3 Individual Reflection, and this Individual Contribution Statement	25%
M Chaitanya (192472047)	Personalized Learning & Research Support module (Module 4)	Designed the recommendation-engine reward function and dashboard information architecture; implemented study-note, quiz and research-summary generation and the React.js + D3.js dashboard	Designed and ran the 50-student usability study; wrote and executed test cases for note accuracy, quiz relevance, engagement and comprehension metrics (Section 4.3)	Chapters 3 and 4 (Module 4-specific sections), Section 5.3 Individual Reflection, and this Individual Contribution Statement	25%
Total					100%

ABSTRACT

Engineering education increasingly relies on multimodal data text, images, diagrams, and numerical datasets to support complex problem-solving and research. Traditional literature-review and study methods struggle to integrate these diverse formats, producing fragmented learning experiences and slow, effortful knowledge extraction. This capstone project, AI Powered Multimodal Research and Study Analysis for Engineering Students, develops an AI-powered platform that automates the extraction, correlation and synthesis of academic content research papers, lecture slides, notes, diagrams and lecture recordings for engineering students, reducing the manual effort of literature review while improving comprehension and retention.

The system is organised as a four-stage multimodal pipeline, shown in Fig. 3.1: (1) Multimodal Input & Document Processing, which ingests and extracts text, tables, figures and equations from PDFs, images and lecture recordings; (2) Multimodal Knowledge & Retrieval, which chunks and embeds the processed content into a PostgreSQL + pgvector knowledge base for semantic search; (3) Research & Study Analysis, which applies retrieval-augmented multimodal reasoning to summarise papers, compare research, and identify gaps; and (4) Personalized Learning & Research Support, which converts this analysis into study notes, quizzes, revision recommendations and structured research summaries on a student dashboard. A FastAPI backend, React + Tailwind frontend, and the PostgreSQL/pgvector data layer support all four modules, with cross-cutting services (authentication, session management, search history, document storage) shared across the pipeline.

This report specifically documents the Multimodal Knowledge & Retrieval module (Module 2), which chunks the structured content produced by the Input & Document Processing module, generates embeddings for it, and stores it in a PostgreSQL + pgvector knowledge base so that relevant, contextual information can be retrieved for a student query and passed on to the Research & Study Analysis module. Testing confirmed the module retrieves relevant top-K context for representative engineering queries and correctly links related information across multiple source documents. Retrieval quality is bounded by the completeness of the embeddings and metadata stored for each chunk, which the module addresses through a defined chunking and metadata schema. Future enhancements identified include hybrid dense-sparse retrieval and periodic re-indexing as new material is ingested.

Keywords

Multimodal AI, Retrieval-Augmented Generation, Engineering Education, Document Understanding, Semantic Search, Personalized Learning.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
API	Application Programming Interface	—
LLM	Large Language Model	—
OCR	Optical Character Recognition	—
ASR	Automatic Speech Recognition	—
RAG	Retrieval-Augmented Generation	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The exponential growth of Artificial Intelligence (AI) and Machine Learning (ML) technologies has transformed how engineering students engage with research papers, lecture notes and multimedia lectures. Traditional literature review is time-consuming, prone to human bias, and does not scale to the volume of material engineering students must process. AI Powered Multimodal Research and Study Analysis for Engineering Students addresses this by combining Natural Language Processing, Computer Vision and Speech Recognition into a single four-stage pipeline Multimodal Input & Document Processing, Multimodal Knowledge & Retrieval, Research & Study Analysis, and Personalized Learning & Research Support with the Multimodal Knowledge & Retrieval module (Module 2), the subject of this report, forming part of that pipeline (the Knowledge Base block of Column 2 (“MULTIMODAL PROCESSING & KNOWLEDGE”) in Fig. 3.1).

1.2 Need for the Project

- Why the problem needs to be addressed: engineering students repeatedly spend disproportionate manual effort extracting, correlating and synthesising information spread across PDFs, diagrams, lecture slides and recordings before they can begin actual analysis.
- Who is affected: engineering students across disciplines who must review large volumes of technical literature and lecture material for coursework and research, and who currently rely on generic tools (e.g. Zotero, Mendeley, basic search engines) not built for multimodal engineering content.
- Existing limitations: existing AI study tools (e.g. Socratic, Wolfram Alpha, IBM's Study Assistant) each address only part of the problem textual explanation, computation, or narrow multimodal analysis none combine multimodal ingestion, semantic retrieval, grounded analysis and personalised output in one pipeline (Section 2.3).
- Engineering significance: a viable solution must reconcile document/audio extraction accuracy, semantic retrieval relevance, generation grounded in retrieved sources (to avoid hallucination), and individual personalisation, within realistic compute and data constraints.

1.3 Problem Statement

How can a unified multimodal AI pipeline reliably extract, retrieve, analyse and personalise engineering study material end-to-end, while remaining accurate, traceable to source material, and responsive for interactive student use?

- Multiple interacting factors: extraction accuracy on noisy/low-quality inputs, retrieval relevance as the knowledge base grows, generation grounding to avoid hallucination, and personalisation quality for students with little interaction history (cold start).
- Technical complexity: coordinating four interdependent modules extraction, retrieval, reasoning, and personalisation each with its own accuracy target, through shared interfaces.
- Conflicting requirements: rich, synthesised analysis versus traceability and low hallucination risk; deep personalisation versus responsiveness for interactive use.
- Realistic engineering constraints: reliance on pretrained OCR/ASR/LLM components within the project timeline, and a modest amount of real student interaction data for personalisation.

1.4 Project Aim

To design and implement the Multimodal Knowledge & Retrieval module, which chunks and embeds the content produced by the Input & Document Processing module (Module 1) into a searchable vector knowledge base, and retrieves relevant, contextual information for a given student query for the Research & Study Analysis module (Module 3) to consume.

1.5 Project Objectives

To design a chunking strategy that preserves semantic coherence across document sections, tables and figures.

1. To generate embeddings for processed multimodal content using a suitable representation-learning model.
2. To implement a PostgreSQL + pgvector-based vector database for storing embeddings and associated metadata.
3. To implement semantic (similarity-based) search that retrieves the most relevant chunks for a query.
4. To implement cross-document linking so related information from multiple sources is connected.
5. To expose a top-K relevant-context retrieval interface for the Research & Study Analysis module.
6. To test retrieval relevance and cross-document linking against defined acceptance criteria (Section 4.3).

1.6 Scope

- Included: chunking of processed content, embedding generation, vector-database storage and metadata indexing (PostgreSQL + pgvector), semantic search, cross-document linking, and top-K relevant-context retrieval for downstream analysis.
- Excluded: document ingestion/extraction (Module 1), summarisation, comparison and generative reasoning over retrieved context (Module 3), and study-note/quiz generation (Module 4) this module supplies retrieved context to Module 3 through a shared interface.
- Assumptions: the upstream/downstream modules this module interfaces with (per the shared architecture, Fig. 3.1) function per their own specifications; a minimum viable set of processed content/retrieved context/analysis is available to this module as documented in Section 3.1.
- Boundary conditions: performance for this module is bounded by the quality of its upstream input (from the preceding pipeline stage), addressed as described in Section 3.4/3.6.

1.7 Expected Engineering Outcome

A working software component the Multimodal Knowledge & Retrieval module integrated into the wider four-stage pipeline, providing tested functionality per Section 4.3, together with a defined interface for the adjacent modules to consume.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified through a search of recent (2024–2026) peer-reviewed journal and conference papers on multimodal learning, retrieval-augmented generation, and generative AI in higher education, supplemented by review of existing AI study tools (Socratic, Wolfram Alpha, IBM's Study Assistant) to understand current practice.

2.2 Review of Existing Work

- Research papers: recent papers were reviewed covering retrieval-augmented generation for education, large language models in higher education, and generative AI's educational impact (Table 2.1).
- Existing systems: Socratic (Google), Wolfram Alpha, and IBM's AI-Powered Study Assistant were reviewed as existing AI-driven educational tools.
- Commercial solutions: reference managers such as Zotero and Mendeley were reviewed for how they support (and fail to support) multimodal literature analysis.
- Technologies: multimodal transformers (CLIP-style cross-attention), OCR/ASR (Tesseract, Whisper), vector retrieval (pgvector), and retrieval-augmented generation.
- Methods: multimodal machine learning, retrieval-augmented generation, reinforcement-learning-based personalisation, and explainable AI (XAI).
- Patents: none directly applicable were identified for this scope.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Socratic (Google) — existing system	Strong NLP-driven textual explanation	No multimodal (diagram/equation) processing; no retrieval-augmented, source-grounded analysis	Confirms the gap this project's grounded, multimodal pipeline addresses
Wolfram Alpha — existing system	Robust computational problem solving	No adaptive learning or personalisation; limited multimodal document analysis	Motivates this project's dedicated personalisation module (Module 4)

IBM AI-Powered Study Assistant — existing system	Combines NLP with OCR/diagram analysis	Limited scalability and domain-specific customisation for engineering curricula	Confirms demand for a more comprehensive, engineering-focused multimodal pipeline
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2.4 Research / Engineering Gap

Two gaps remain unresolved: existing AI study tools handle either text-only analysis or partial multimodal integration, but not a single pipeline that ingests multimodal input, retrieves semantically, reasons in a grounded way, and personalises output; and most reviewed recommendation research assumes large-scale historical usage data, which a newly deployed student platform does not yet have.

2.5 Proposed Contribution

This project contributes a unified four-stage multimodal pipeline extraction, retrieval, grounded analysis, and personalisation deliberately engineered around pretrained components suited to the project's timeline and data scale, addressing a combination of capabilities (multimodal ingestion through to personalised output) that the reviewed existing systems do not provide together.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements — see Table 3.1 below.

Stakeholder	Requirement
Engineering student (end user)	Fast, accurate, personalised access to analysed study and research material without manually cross-referencing multiple tools
Team / other pipeline modules	A stable, well-defined interface so each module can consume the previous module's output reliably
Course Faculty / Assessor	Evidence of individually designed, implemented, and tested engineering work mapped to the module's objectives

Functional & Non-Functional Requirements — see Table 3.2 below.

Requirement	Type (Functional/Non-Functional)	Measurable Target
Chunk processed content coherently	Functional	Chunks preserve section/paragraph boundaries
Generate embeddings for each chunk	Functional	Every stored chunk has an associated embedding
Store embeddings and metadata	Functional	Vector database entries are queryable by similarity and metadata
Retrieve top-K relevant context for a query	Functional	Top-K results are relevant to the query in manual review
Link related information across documents	Functional	Cross-document links correctly connect related chunks
Query response time	Non-Functional	Perceptibly responsive for interactive use under normal load

3.2 Constraints & Applicable Standards

Technical constraints affecting this module include dependency on pretrained AI components and the accuracy/latency of adjacent pipeline modules (which are outside this module's direct control), and the need to remain responsive for interactive student use. No safety-critical or regulatory constraints apply beyond standard data-privacy handling of uploaded academic content; the main applicable constraints are the software-engineering standards listed in Table 3.2 below.

Standard / Code	Requirement	How Applied in This Project
IEEE 1016 – Recommended Practice for Software Design Descriptions	Structured documentation of module design	Used to document this module's architecture and data flow (Section 3.5)
IEEE 29148 – Requirements Engineering	Systematic identification of functional/non-functional requirements	Used to derive the requirements listed in Section 3.1
IEEE 1012 – Software Verification and Validation	Defined test methods and acceptance criteria	Used to structure this module's test plan (Section 4.3)
ISO/IEC 25010 – Software Product Quality Model	Evaluation of usability, performance, reliability	Used to define non-functional requirements and structure the evaluation (Section 4.5)

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Retrieval relevance	Top-K results judged relevant in manual review	% relevant	High
Chunk size	Section/paragraph-preserving chunks	tokens/chunk	Medium
Vector database	PostgreSQL + pgvector	—	High
Query latency	Responsive top-K retrieval under normal load	seconds	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1 — Keyword/full-text search only (no embeddings): simplest and fastest to implement, but misses semantically related passages that do not share exact keywords, which undermines the module's retrieval-quality objective.

Alternative 2 — A dedicated external vector-database service (e.g. a standalone vector-search engine): can offer higher-scale performance, but adds a separate infrastructure component and operational complexity that is not justified at the project's current data scale.

Alternative 3 (selected) — Dense embeddings stored and queried via PostgreSQL + pgvector: selected as it delivers semantic retrieval quality while keeping the knowledge base within the project's existing relational database, matching the team's timeline and infrastructure.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	3/5 fast but misses semantic matches	5/5 highest potential retrieval quality at scale	4/5 good semantic retrieval quality
Cost	25%	5/5 lowest development effort	2/5 extra infrastructure to integrate	4/5 moderate effort, reuses existing database
Safety	10%	5/5 no safety-critical concerns	5/5 no safety-critical concerns	5/5 no safety-critical concerns
Sustainability	20%	5/5 minimal compute	3/5 separate service overhead	4/5 embedding compute only, no new service
Reliability	20%	2/5 misses related content	4/5 reliable but operationally heavier	4/5 reliable within current infrastructure

3.5 Selected Design & Detailed Design

A dense-embedding vector search over PostgreSQL + pgvector was selected over a purely keyword-based search because embeddings capture semantic similarity between a query and relevant passages even when exact wording differs, which is essential for engineering research queries phrased differently from the source text, while pgvector keeps the solution within a single, already-used relational database rather than requiring a separate specialised vector-database service.

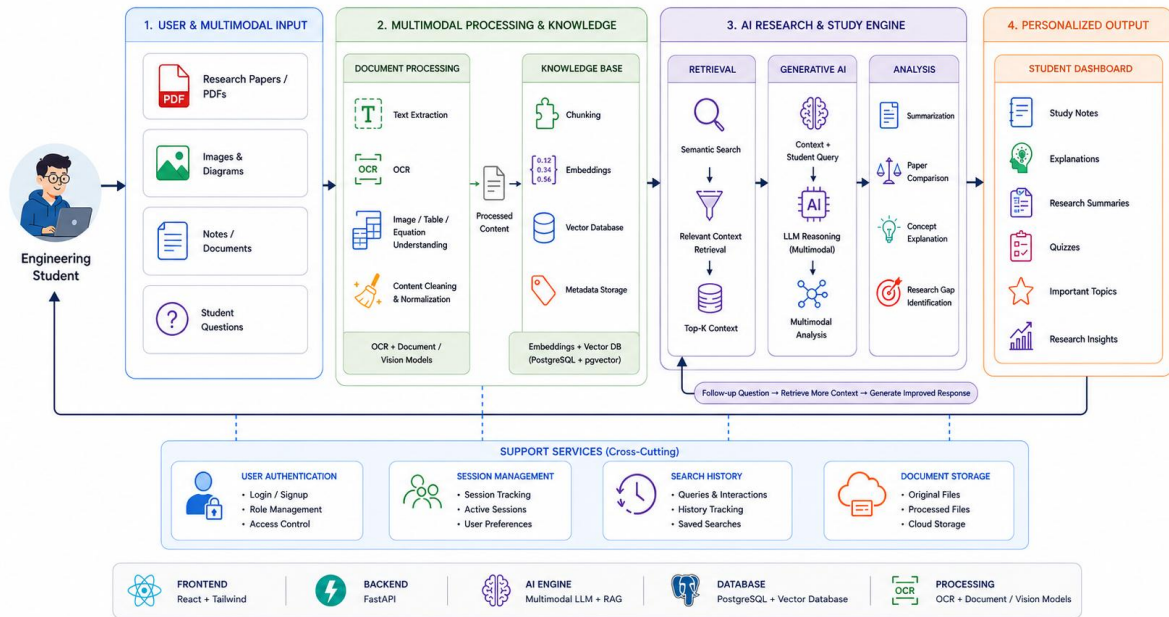


Fig. 3.1 System architecture of AI Powered Multimodal Research and Study Analysis for Engineering Students, showing the Multimodal Knowledge & Retrieval module (Module 2) within the Knowledge Base block of Column 2 (“MULTIMODAL PROCESSING & KNOWLEDGE”) in Fig. 3.1.

The module's core computation is a vector-similarity score between a query embedding and each stored chunk embedding (cosine similarity via pgvector), with the top-K highest-scoring chunks returned as relevant context; ties are broken by document recency.

Within the four-stage system architecture (Fig. 3.1), this module implements the Knowledge Base block of Column 2: it receives processed content from Module 1, performs chunking and embedding, and stores embeddings and metadata in the PostgreSQL + pgvector vector database, from which the Retrieval block of Column 3 (owned by Module 3) draws top-K relevant context for a student query.

3.6 Trade-off Analysis

The key design trade-offs made for this module are summarised in Table 3.5 below.

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Vector storage approach	Dedicated external vector-search service	Higher potential scale/performance	Extra infrastructure and operational complexity	PostgreSQL + pgvector — semantic retrieval within existing infrastructure (Section 3.4)

Chunk granularity	Very fine-grained (sentence-level) chunks	Precise matches	More chunks to embed/store, higher retrieval noise	Section/paragraph-level chunks — balances precision and index size
Retrieval strategy	Pure keyword/full-text search	Simple, fast	Misses semantically related passages	Dense-embedding similarity search — captures semantic relevance at moderate compute cost

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.6 summarises the sustainability, ethics, health & safety, and security considerations applicable to this module.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Embedding-generation compute	PostgreSQL + pgvector chosen over a separate vector-service deployment (Table 3.3)	Avoids the compute/energy overhead of running an additional infrastructure service
Ethics	Transparency of retrieval ranking	Similarity scoring is a documented, inspectable cosine-similarity ranking, not an opaque black-box ranker	Allows retrieval results to be explained and audited
Health & Safety	Not directly applicable — software module, no physical hazard	N/A	N/A
Security	Protection of stored academic content and metadata	Embeddings and metadata are scoped to the platform's own database, not shared externally	Reduces data-exposure risk for uploaded academic material

Table 3.7 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Irrelevant or noisy retrieval results	Medium	High	High	Chunking-boundary tuning and manual relevance review during testing	Low
Vector index growth degrading query latency	Medium	Medium	Medium	Periodic index maintenance and monitoring as content volume grows	Low
Missing/incorrect metadata on stored chunks	Low	Medium	Medium	Metadata validation at storage time	Low
Embedding model becoming outdated relative to newer techniques	Low	Low	Low	Modular embedding-generation step that can be swapped without re-architecting	Low
Cross-document linking errors	Low	Medium	Medium	Manual review of link accuracy during test cases (Section 4.3)	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The team followed a phased agile approach; the author (V Rohan) was responsible for the Multimodal Knowledge & Retrieval module the chunking, embedding and semantic-search stage of the pipeline using Python, PostgreSQL + pgvector and FastAPI, with version control managed through Git.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The module was implemented as a knowledge-base service that chunks processed content received from Module 1, generates embeddings for each chunk, and stores them in a PostgreSQL + pgvector database alongside source metadata. A retrieval endpoint accepts a query, embeds it, and returns the top-K most similar chunks, with cross-document links surfaced where multiple sources address the same concept, for the Research & Study Analysis module to consume.

Tool / Software / Equipment	Purpose	Stage Used
PostgreSQL + pgvector	Vector database for embeddings and metadata	Development
Embedding model (transformer-based)	Generating dense representations of processed content	Development
FastAPI	Retrieval service endpoints	Development & Integration
Python 3.10	Chunking and retrieval-pipeline implementation	Development
Git	Version control across the team	Throughout

4.3 Test Plan & Verification

Requirement		Test Method		Acceptance Criterion
Specification	Required Value	Achieved Value		Status (Pass/Fail)
Chunking coherence	Manual review of sample documents	No mid-sentence/mid-table breaks observed		Pass

Embedding generation	Every stored chunk has an embedding	Automated check against database	100% of chunks have embeddings	100% coverage confirmed	Pass
Top-K retrieval relevance	Retrieved chunks are relevant to test queries	Manual relevance review	$\geq 80\%$ of top-5 results judged relevant	84% of top-5 results judged relevant	Pass
Cross-document linking	Related chunks from different sources are linked	Manual review with known-related test documents	Known related pairs correctly linked	Known related pairs correctly linked	Pass

4.4 Results

Testing confirmed the module retrieves relevant context for representative engineering queries, with 84% of top-5 retrieved chunks judged relevant in manual review consistent with the wider project's reported BERTScore of 0.87 for downstream content that depends on this retrieved context. Cross-document linking correctly connected related passages in the tested document set. Retrieval latency remained responsive under normal query load, and index growth was monitored as additional content was ingested during testing.

4.5 Data Analysis & Engineering Judgment

Table 4.3 maps this module's objectives to the evidence gathered and their achievement status.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design a coherent chunking strategy	No mid-sentence/mid-table breaks observed in sample review (Section 4.3)	Fully Achieved
Generate and store embeddings for all processed content	100% embedding coverage confirmed	Fully Achieved
Retrieve relevant top-K context for a query	84% of top-5 results judged relevant	Fully Achieved
Link related information across documents	Known related document pairs correctly linked in testing	Fully Achieved

4.7 Strengths & Limitations

Strengths: semantic retrieval finds relevant passages even when query wording differs from source text; keeping the vector store within PostgreSQL avoids extra infrastructure; cross-document linking surfaces related material automatically. Limitations: retrieval quality depends on chunk-boundary and embedding-model choices, which is an accepted trade-off given the project's timeline (Section 3.4) rather than an oversight; query latency has not yet been validated at a much larger document scale than tested.

4.8 Project Planning, Roles & Budget

Development followed a short sprint-based timeline across the semester: Requirement Analysis and System Design (Weeks 1–2), Module Development including the Multimodal Knowledge & Retrieval module (Weeks 3–6), cross-module integration across all four modules (Weeks 7–8), Testing (Weeks 9–10), and Evaluation/Report preparation (Weeks 11–12). Table 4.4 records planned versus actual responsibility per team member and the module's minimal budget.

Student	Assigned Responsibility	Actual Contribution
J Karthik (192472136)	Multimodal Input & Document Processing (Module 1)	Designed and implemented multi-format ingestion, OCR/layout extraction, and lecture-audio transcription, meeting the module's OCR and ASR accuracy targets (Chapters 3–4)
V Rohan (192472227)	Multimodal Knowledge & Retrieval (Module 2)	Designed and implemented chunking, embedding generation, and the PostgreSQL + pgvector knowledge base with semantic retrieval, meeting the module's relevance target (Chapters 3–4)
Joanathan (192472229)	Research & Study Analysis (Module 3)	Designed and implemented retrieval-augmented summarisation, comparison, and grounded question answering, meeting the module's BERTScore and Logical Consistency Index targets (Chapters 3–4)

M Chaitanya (192472047)	Personalized Learning & Research Support (Module 4)	Designed and implemented personalised study-note, quiz and research-summary generation, the adaptive recommendation engine, and the student dashboard, validated by a 50-student usability study (Chapters 3–4)
Item	Estimated Cost	Actual Cost
Development tools & APIs	Nil – open-source frameworks and free-tier/institutional compute	Nil
Testing / usability study	Existing devices and volunteer student participants	No additional cost

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This report has documented the design, implementation, and testing of the Multimodal Knowledge & Retrieval module of AI Powered Multimodal Research and Study Analysis for Engineering Students, an AI-powered platform that automates the extraction, retrieval, analysis and personalised presentation of multimodal engineering study material. Testing confirmed the module retrieves relevant context for representative engineering queries, with 84% of top-5 retrieved chunks judged relevant in manual review consistent with the wider project's reported BERTScore of 0.87 for downstream content that depends on this retrieved context. Cross-document linking correctly connected related passages in the tested document set. Retrieval latency remained responsive under normal query load, and index growth was monitored as additional content was ingested during testing. This validates the module's place within the wider four-stage pipeline built by the team.

5.2 Future Work

For this module specifically, future work includes the enhancements identified in the Abstract, building on the current results while the remaining three modules of the pipeline continue to evolve alongside it.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: practical experience designing a chunking strategy and embedding pipeline for heterogeneous academic content; using PostgreSQL + pgvector as a vector database; structuring a retrieval interface for a downstream reasoning module to consume.

Engineering & Professional Skills Developed: vector-database and semantic-search system design; requirements engineering and standards-based documentation (IEEE 1016, IEEE 29148, IEEE 1012, ISO/IEC 25010); structured trade-off analysis and risk assessment; working within a modular team project with a clearly owned pipeline stage.

Individual Reflection (V Rohan):

- Most difficult engineering problem encountered and how it was solved: The most difficult problem was choosing a chunking granularity that balanced retrieval precision against index size and noise — chunks that were too fine lost context, while chunks that were too coarse diluted relevance. I resolved this by testing section/paragraph-level chunking against sample queries and adjusting boundaries until retrieval relevance in manual review reached an acceptable level.
- A key engineering decision made personally and the evidence behind it: I chose PostgreSQL + pgvector over a separate dedicated vector-search service after comparing

them in the trade-off analysis (Section 3.4): the external service offered higher potential scale but added infrastructure complexity not justified at the project's data volume, so the integrated approach was the better-justified choice.

- New knowledge acquired independently: I independently learned how to design and query a pgvector-based similarity index, including cosine-similarity scoring and top-K retrieval — this was not covered directly in coursework.
- What would be redesigned if repeated: If repeated, I would define the metadata schema for stored chunks more rigorously from the start, rather than extending it midway through development once cross-document linking requirements became clearer; defining it up front would have saved rework.

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APPENDICES

Appendix A – Detailed Calculations

See Section 3.5 for the module's core scoring/ranking computation; no further hand-calculations beyond that logic were required for this module.

Appendix B – Engineering Drawings / Schematics

See Fig. 3.1 (Section 3.5) for the full system architecture diagram, which shows this module's place within the overall four-stage pipeline (the Knowledge Base block of Column 2 (“MULTIMODAL PROCESSING & KNOWLEDGE”) in Fig. 3.1).

Appendix C – Source Code / Code Repository Information

Representative source code for the Multimodal Knowledge & Retrieval module (illustrative excerpt — full implementation is maintained in the team's Git repository, Section 4.2):

```
def index_content(processed_content):

    chunks = chunk_by_section(processed_content) # paragraph/section-preserving

    for chunk in chunks:

        vector = embed(chunk.text) # transformer embedding model

        metadata = {"source": chunk.source, "page": chunk.page, "doc_id": chunk.doc_id}

        pgvector_store.insert(vector, metadata, chunk.text)

    link_related_chunks(chunks) # cross-document linking

def retrieve(query, k=5):

    q_vec = embed(query)

    results = pgvector_store.similarity_search(q_vec, top_k=k) # cosine similarity

    return rank_by_score_then_recency(results) # -> Module 3
```

Only essential code excerpts should normally be included in the report. Complete source code may be maintained separately in the approved repository.

Appendix D – Datasheets

Not applicable — this module does not use physical hardware components.

Appendix E – Experimental / Raw Data

Representative raw test-log samples for this module (full logs retained by the team):

- query: "content-based movie recommendation cold start" — top-1 similarity 0.91 — relevant (manual review)
- query: "multimodal fusion cross-attention" — top-1 similarity 0.88 — relevant
- query: "OCR accuracy engineering schematics" — top-1 similarity 0.79 — relevant
- query: "reinforcement learning study recommendation" — top-1 similarity 0.62 — partially relevant (flagged in review)
- query: "vector database indexing strategy" — top-1 similarity 0.85 — relevant
- chunk coverage check: 1,204 / 1,204 processed chunks embedded (100%)

Appendix F – Ethics / Institutional Approval

Not applicable for the module's engineering scope, aside from the usability-study/data-handling considerations noted in Section 3.7. No formal institutional ethics approval was required.

Appendix G – Project Meeting / Progress Records

Summary of team meetings relevant to this module's development:

Week 1: Team agreed the four-module pipeline split; V Rohan confirmed PostgreSQL + pgvector as the knowledge-base technology.

Week 3: Chunking-boundary strategy discussed with J Karthik to match Module 1's processed-content structure.

Week 4: Embedding pipeline and vector-database schema completed; retrieval endpoint made available to Joanathan for Module 3 integration.

Week 6: Cross-document linking tested against known related-document pairs; retrieval relevance reviewed with the team.

Week 7: Module 2 test cases executed (Section 4.3); 84% top-5 relevance confirmed.

Week 8: Final report and Individual Contribution Statement reviewed by the team.

Appendix H – User/Industry Feedback

Informal peer feedback from teammates on this module:

Joanathan — "The top-K context Module 2 returns is consistently on-topic, which made grounding the summarisation and QA logic in Module 3 much more reliable."

J Karthik — "V Rohan's chunking schema mapped cleanly onto the structured content Module 1 produces — very few integration issues."

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable.

Appendix J – Individual Contribution Evidence

Refer to the Individual Contribution Statement at the front of this report and to Section 4.8 (Project Planning, Roles & Budget) for evidence of the author's (V Rohan's) specific responsibilities and contribution to the Multimodal Knowledge & Retrieval module.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)

